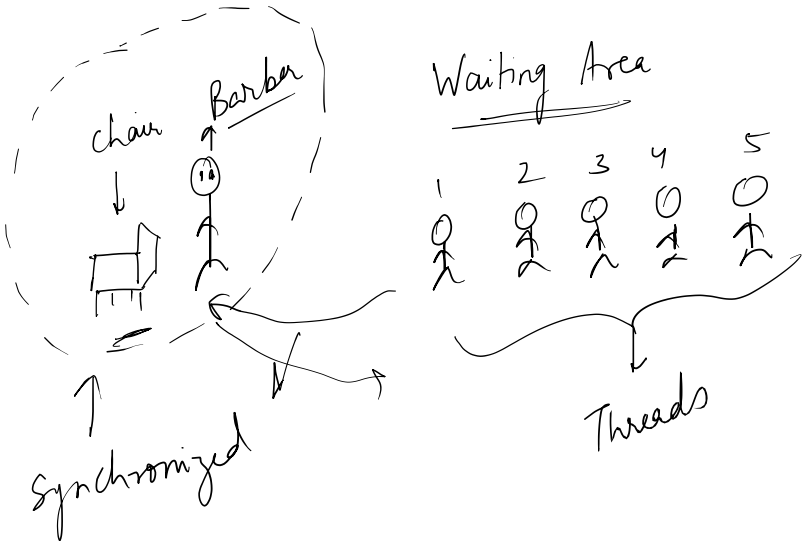
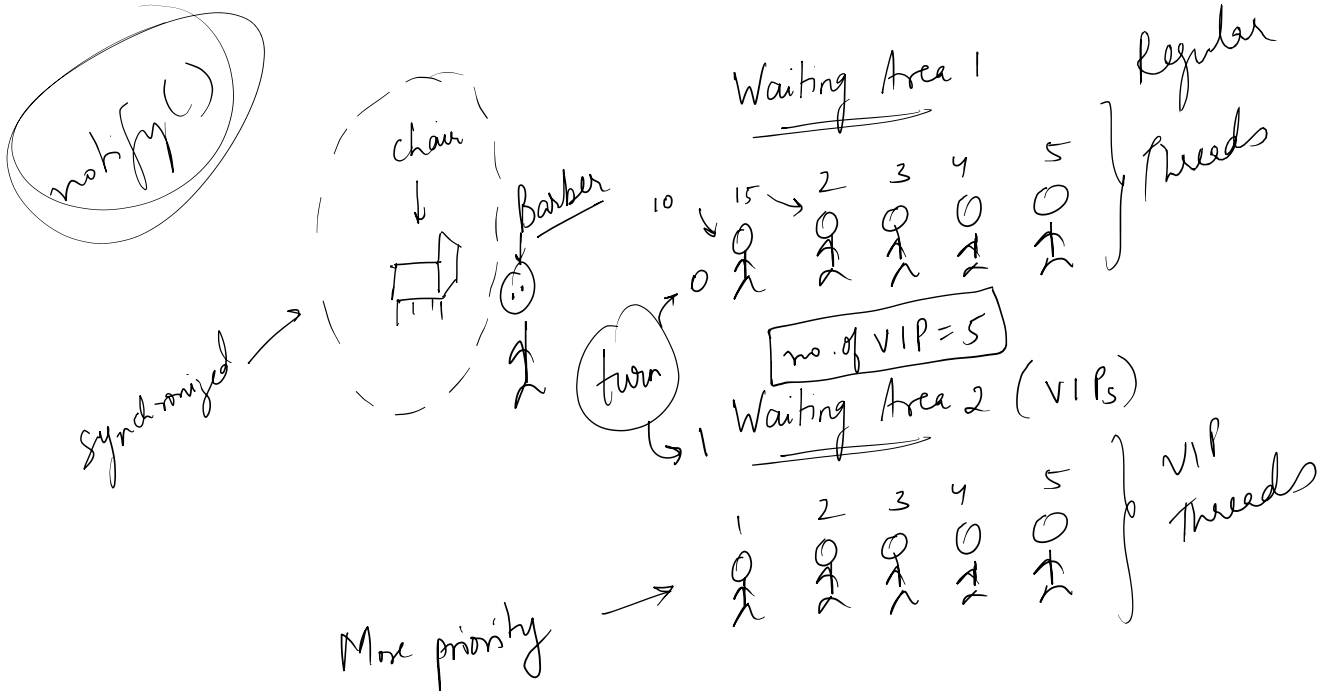
Re-entrant Lock & Condition

Barber Shop





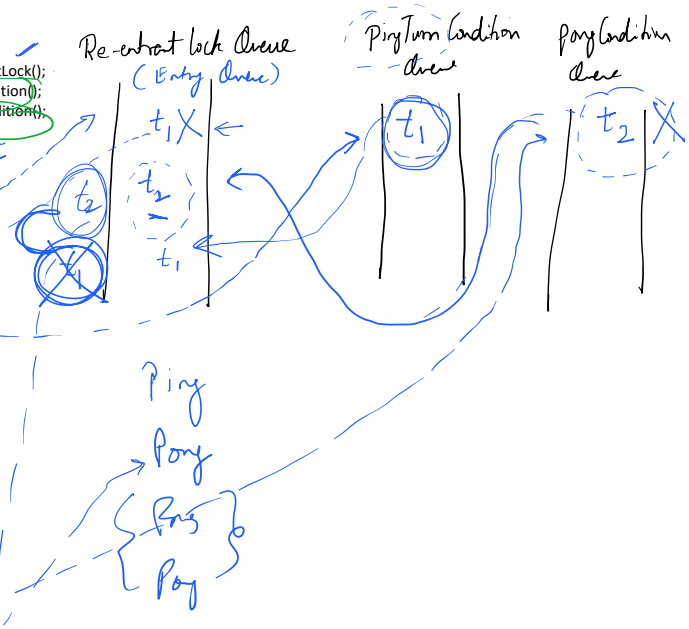
It's like
monitor object

Conditions
Queue 1 (Condition 1)
$T_1, T_2, T_3, T_4,$

Re-entrant lock → [illegible] on which you acquire a lock

Queue 2 (Condition 2) T_5, T_6, T_7

Queue 3 (Condition 3) $T_8, T_9, T_{10}, T_{11}, T_{12}$

Queue 4 (Condition 4) $T_{13}, T_{14}, T_{15}, T_{16}$

```
public class PingPong {
  private final ReentrantLock lock = new ReentrantLock();
  private final Condition pingTurn = lock.newCondition();
  private final Condition pongTurn = lock.newCondition();
  private boolean isPingTurn = true;

  public void ping() {
    for (int i = 0; i < 5; i++) {
      lock.lock();
      try {
        while (!isPingTurn) {
          pingTurn.await();
        }
        System.out.println("Ping");
        isPingTurn = false;
        pongTurn.signal();
      } catch (InterruptedException e) {
        Thread.currentThread().interrupt();
      } finally {
        lock.unlock();
      }
    }
  }

  public void pong() {
    for (int i = 0; i < 5; i++) {
      lock.lock();
      try {
        while (isPingTurn) {
          pongTurn.await();
        }
        System.out.println("Pong");
        isPingTurn = true;
        pingTurn.signal();
      } catch (InterruptedException e) {
        Thread.currentThread().interrupt();
      } finally {
        lock.unlock();
      }
    }
  }

  public static void main(String[] args) {
    PingPong pp = new PingPong();

    Thread t1 = new Thread(pp.ping);
    Thread t2 = new Thread(pp.pong);

    t1.start();
    t2.start();
  }
}
```